

ARYAN LOKESH

Graduate Software and AI Engineer

United Kingdom | aryanlokesh.me | aryan.lokesh4@gmail.com | 07833806683
linkedin.com/in/aryan-l-4516562b4 | github.com/Aio777

PROFILE

MComp Computer Science with Artificial Intelligence graduate with project experience in Android development, Rails applications, audio machine learning, GPU programming and numerical computing. UK Graduate visa holder.

EDUCATION

MComp Computer Science with Artificial Intelligence | University of Sheffield | 2022-2026

Four-year integrated master's, 2:1. Final-year average: 70.1%. Selected results: Parallel Computing with GPUs 79; Darwin Project 75; Speech Technology 74.

Cambridge International A Levels (June 2021): Information Technology A*, Mathematics A, Chemistry B, Physics B; English Language B (AS Level).

TECHNICAL SKILLS

Languages: Python, C/C++, Kotlin, Ruby, SQL

Frameworks and tools: PyTorch, Ruby on Rails, PostgreSQL, CUDA/OpenMP, Jetpack Compose, Room, NumPy/CuPy, SciPy, Git, Linux

SELECTED UNIVERSITY PROJECTS

Productiv - Rails Productivity Platform | 2025-2026

Ruby on Rails 8, PostgreSQL | Genesys team project

- Year 4 team project: a productivity platform with personal/shared tasks, team projects, Pomodoro, notifications, gamification and role-based analytics.
- The team codebase includes RSpec/Capybara tests, performance examples and CI configuration for testing, security and linting.

ReCoLA - Android Property Management | 2025

Kotlin, Jetpack Compose, Room, Google Maps | Six-person team

- Implemented property creation, viewing, editing and deletion, maps/geocoding, proximity sorting, supplier assignment, internal image handling and related tests.
- Connected Compose screens to Room persistence and map/list navigation, with alphabetical sorting when live location was unavailable.

TB-Turing / HeAR - Audio-Based TB Screening Research | 2025-2026

Python, PyTorch, Google HeAR, HPC | HeAR workstream within team research

- Developed patient-level multiple-instance learning over frozen HeAR cough embeddings, with grouped five-fold cross-validation, sensitivity-targeted thresholds and bootstrap analysis.
- The DS2 symptom-assisted configuration reached 0.830 AUROC in retrospective internal cross-validation; this was screening research, not clinical validation.

Parallel Computing with OpenMP and CUDA | 2026

C/C++, CUDA, OpenMP, Nsight | Individual coursework

- Implemented and optimised Count Gliders, Histogram and Emboss in OpenMP and CUDA against supplied serial CPU references.
- Checked correctness and profiled execution with Nsight; compared end-to-end timings including allocation, host-device transfers, kernel execution and cleanup.

Heston Option Pricing and Calibration | 2024-2025

Python, NumPy/CuPy, PyTorch, SciPy | Individual dissertation

- Implemented CPU COS pricing, batched CuPy GPU pricing and bounded L-BFGS-B calibration of the Heston stochastic-volatility model to filtered SPX option contracts.
- Trained a ten-feature PyTorch surrogate on 600,000 synthetic samples and compared calibration runtime and pricing error across CPU, GPU and surrogate paths.

Speech Technology - Neural VAD and GMM Modelling | 2026

Python, PyTorch, scikit-learn | Individual coursework

- Compared MLP, recurrent and temporal-convolutional voice-activity detectors; the selected TCN achieved 7.73% EER over 495,486 cross-corpus test frames.
- Separately implemented Gaussian mixture models for vowel classification and closed-set speaker identification, comparing mixture complexity and utterance-level decisions.